Lab. 2 – Numerical analysis of MTLs

Assigned: **Thursday March 26th, 2026**
Due: **Friday April 17th, 2026 at class starting**

Problem nr. 1

Using microstrip technology and alumina dielectric material (ε_r = 9.8, thickness h = 1.6 mm), design a microstrip line with equivalent characteristic impedance of 50 Ω.
Assume that the total voltage and current in the input section of the line (z = 0) are equal to $V_{iso}(0)$ = 10 mV and $I_{iso}(0)$ = 0.4 mA, respectively; write a Matlab script to evaluate the voltage $V_{iso}(z)$ along the line at the frequency f = 10 GHz, and plot it for $0 \leq z \leq L$ with L = 100 mm.

Problem nr. 2

Consider then the case in which close to this line there is another, equal to the first one, and that the gap between the two line is s. The voltage and the current in the input section of the first conductor are equal to those on the single microstrip, i.e. $V_1(0) = V_{iso}(0)$ = 10 mV and $I_1(0) = I_{iso}(0)$ = 0.4 mA, while on the second conductor $V_2(0)$ = 0 and $I_2(0)$ = 0.
Assume that s = 0.5 mm: Using the application

http://wcalc.sourceforge.net/cgi-bin/coupled_microstrip.cgi

compute the characteristic parameters (Z_{even}, Z_{odd}, k_{even}, k_{odd}) of the modal line.
Plot $|V_1(z)|$ and $|V_2(z)|$ along the two physical lines for $0 \leq z \leq L$. Compare the resulting plots with that for $|V_{iso}(z)|$.

Repeat the evaluation of the characteristic parameters (Z_{even}, Z_{odd}, k_{even}, k_{odd}) of the modal line and plot $|V_1(z)|$ and $|V_2(z)|$ along the two physical lines for the case in which s = 10 mm.